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Current Affiliation: Harbin Institute of Technology, Shenzhen, China.

Research Areas: Green Hydrogen Economics; Energy Transition; Input–Output Analysis; Machine Learning Forecasting; Low-Carbon Energy Policy



Academic Profile

Master’s student in Applied Economics at Harbin Institute of Technology, Shenzhen, with research focused on green hydrogen penetration dynamics, electricity demand forecasting, and macroeconomic impacts in developing energy systems. My current thesis examines Bangladesh’s power sector using Bass diffusion modeling and input–output analysis, supported by machine learning-based electricity demand forecasting, fuel displacement analysis, avoided emissions accounting, and techno-economic cost assessment. I am particularly interested in low-carbon energy transition, renewable hydrogen deployment, energy security, and economy-wide impacts of clean energy technologies in developing and emerging economies.

Education

Harbin Institute of Technology, Shenzhen, China Doctor of Philosophy (Ph.D.) in Management Science and Engineering Aug 2026 to Present (Continue) Master’s Degree in Applied Economics Oct 2024 - June 2026	
Eastern University, Bangladesh Bachelor’s Degree Electrical and Electronic Engineering Year: 2016-2019	Namajgarh Gausul Azam Kamil Madrasah, Bangladesh Bachelor’s and Master’s Degree Political Science Year: 2010-2013, and 2013-2014

Research Experience

Graduate Researcher, Harbin Institute of Technology, Shenzhen, China.

2024 to Present

Master’s Thesis Research

Title: Green Hydrogen Penetration Dynamics and Macroeconomic Impacts in Bangladesh’s Power Sector: A Bass Diffusion and Input–Output Approach

Supervisor: Associate Professor Chen Xiaobei.

This research investigates how green hydrogen could penetrate Bangladesh’s fossil-fuel dependent power sector and what macroeconomic effects may arise from such a transition. The study uses a dual baseline electricity demand framework, comparing independent model-based forecasts with the official IEPMP 2023 baseline. Bass diffusion modeling is applied to estimate low, moderate, and high green hydrogen penetration pathways, while input–output analysis is used to evaluate economy wide impacts.

Key components include:

- Electricity demand forecasting using ARIMA, LSTM, and ensemble models.
- Bass diffusion modeling for green hydrogen penetration under data scarcity.
- Fuel displacement analysis for natural gas, coal, and oil-based generation.
- Avoided carbon emissions estimation under different hydrogen penetration scenarios.
- Techno-economic assessment of green hydrogen cost parity.
- Input-output analysis of macroeconomic effects, gross output and value-added impacts.

Key results show that green hydrogen penetration could reach 9.87%, 19.99%, and 30.00% by 2050 under low, moderate, and high scenarios, generating about 43.62 TWh, 88.36 TWh, and 132.61 TWh under the LSTM-based baseline. The study also finds significant fossil fuel displacement, avoided emissions potential, and green hydrogen cost parity around 2047. Input–Output results further indicate that GH2-related infrastructure investment could generate about USD 12.4 billion in additional gross output by 2050, around 1.8% of projected GDP, while supporting domestic value-added creation and reducing fossil fuel import dependence.

Research Interests

- Green hydrogen economics and policy analysis
- Renewable energy transition, Development, Deployment & Life-Cycle analysis.
- Energy demand forecasting & Modeling.
- Bass diffusion, New technology adoption, Diffusion & Expansion modeling.
- Input–Output analysis, Economic impact assessment & Import dependence analysis.
- Fossil Fuel & Energy security Modeling.
- Machine learning applications in energy Economics and Environmental Science.
- Climate policy and low-carbon development.

Manuscripts in Preparation

- Muhibullah MD. Green Hydrogen Penetration under Data Scarcity: A Bass Diffusion Application to Bangladesh's Power Sector. Manuscript in preparation.
- Muhibullah MD. Economy-Wide Impacts of Green Hydrogen Deployment in a Fossil-Fuel-Dependent Power Sector: An Input–Output Approach. Manuscript in preparation.
- Muhibullah MD, Lan Rui and Chen Xiaobei. Carbon Quota Price Forecasting in Chinese and European Carbon Markets Using Machine Learning Approaches. Manuscript in preparation.

Technical Skills

Programming and Data Analysis: Python, C++, Java, HTML, CSS, MySQL, Excel, Stata.

Machine Learning and Forecasting: LSTM, ARIMA, Random Forest, Naive, Ensemble Forecasting, Time-Series Backtesting, Model Validation, Forecast Accuracy Metrics Analysis.

Economic Modeling: Input–Output Analysis, Scenario Analysis, Cost-Benefit and Policy Impact Assessment, Energy Demand Modeling, Fuel Displacement Analysis, Emission Accounting.

Data Visualization and Academic Tools: Matplotlib, Excel Charts, Power BI, Tableau, Canva, Academic Tables, Research Figures, LaTeX, Zotero, Endnote, Microsoft Word, PowerPoint.

Hardware, Embedded Systems, and Web Integration: AVR, STM, PLC, Arduino, Raspberry Pi, Analog & Digital Sensors, Microcontrollers, MCU Programming, ROS, Data Reading and Writing, Embedded System Design, Hardware-Software Integration, Web Integration, IOT & GUI Integration.

Research Methods: Quantitative Research, Literature Review, Econometric Modeling, Time-Series Analysis, Energy Scenario Modeling, Policy Analysis, Baseline Analysis, Feasibility Assessment.

Academic Strengths

- Strong interdisciplinary background combining applied economics, energy systems, engineering, machine learning, and low-carbon policy analysis.
- Experience in machine learning and econometric methods to energy transition modeling.
- Research focus on developing economies and fossil-fuel-import-dependent energy systems.
- Ability to develop with long-term energy transition scenarios, official policy baselines, quantitative modeling frameworks and data-driven empirical validation methods.
- Prepared to develop thesis-based research into journal publications and conference papers.

Professional and Practical Experience

Founder & CEO / Technical Developer / Project Manager

Bangladesh (2014–2023)

- Teaching and working experience on engineering, hardware, robotics, & tech project development.
- Practical experience in electronics, programming, project management, & applied problem-solving.
- Collaborated with universities, students, and industrial clients on technical & educational projects.
- Built strong independent learning capacity in Robotics, Coding, & applied technology development.

Languages

- Chinese: Basic and Conversational
- Bengali: Native
- English: Advanced academic working proficiency
- Hindi/Urdu: Conversational

References

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Associate Professor Ji Junping

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